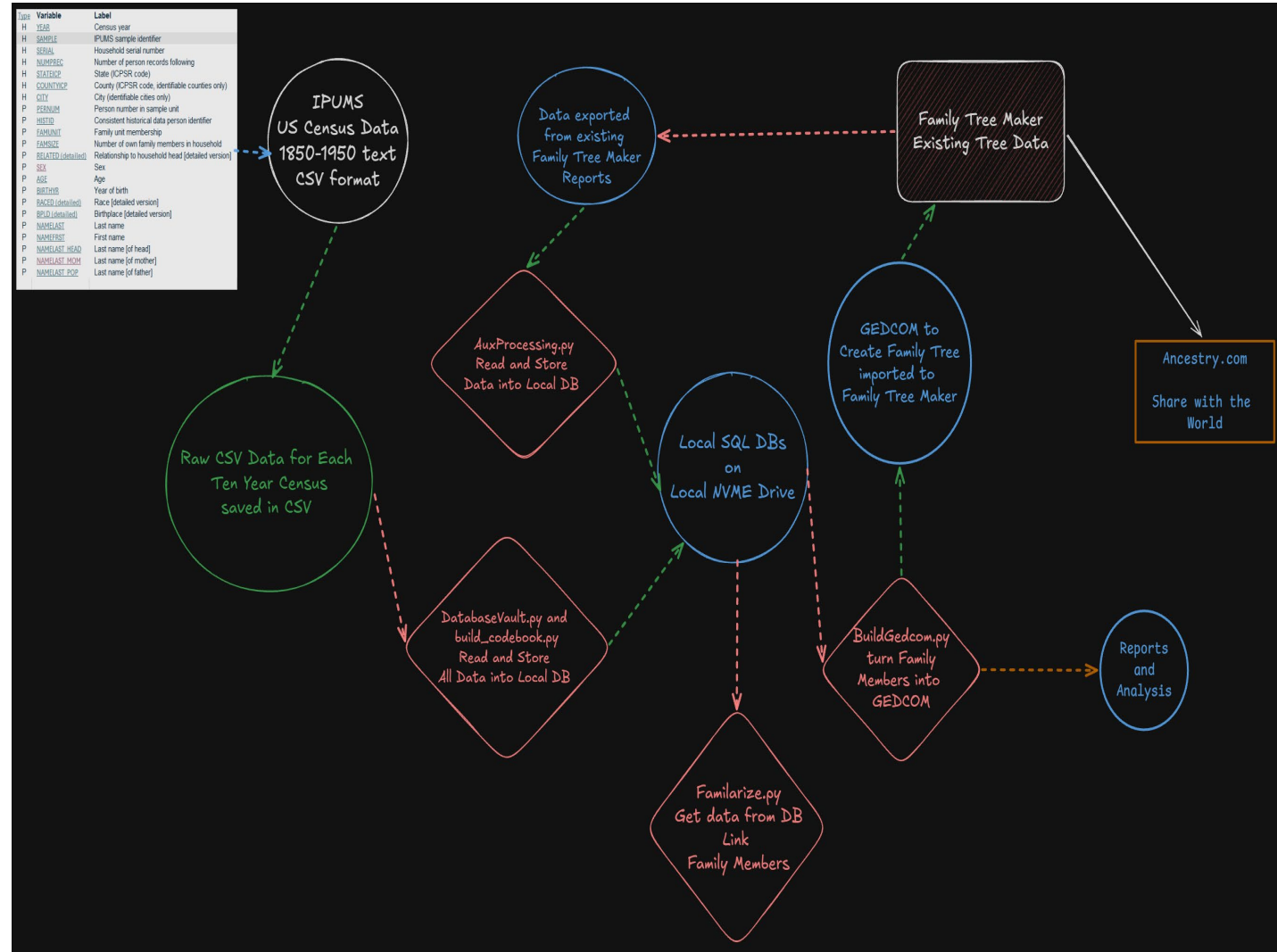


Data Flow from IPUMS to GEDCOM

Create GEDCOM(s) for Family History Programs based on Available
Census Data

Data From IPUMS to Ancestry and Back

1. Download census data from IPUMS.
2. Download Codebook data from IPUMS.
 1. States
 2. Counties
 - BPL : {
 - description :Birthplace [general version] ,
 - codes : {
 - 001 :Alabama ,
 - 002 :Alaska ,
 - 004 :Arizona ,
 - 005 :Arkansas ,
 - 006 :California ,
 - 008 :Colorado ,
 - 009 :Connecticut ,
3. Build SQLite database(s) with census data.
4. Query DBs for info to collate family trees.
5. Build separate DB from existing Ancestry.com family trees.
6. Merge the census and family tree data.
7. Generate a new GEDCOM file to use in Ancestry.comtree or other application.



IPUMS Selected Variables

- Some are automatics from IPUMS
- You decide which are in the DB
- BASIC.txt output shows what is returned from IPUMS processing

Type	Variable	Label
H	YEAR	Census year
H	SAMPLE	IPUMS sample identifier
H	SERIAL	Household serial number
H	HHWT	Household weight
H	CLUSTER	Household cluster for variance estimation
H	STRATA	Household strata for variance estimation
H	GQ	Group quarters status
H	NUMPREC	Number of person records following
H	HHTYPE	Household Type
H	STATEICP	State (ICPSR code)
H	COUNTYICP	County (ICPSR code, identifiable counties only)
H	METAREA (general)	Metropolitan area (pre-2013 delineations, identifiable areas only) [general version]
H	METAREAD (detailed)	Metropolitan area (pre-2013 delineations, identifiable areas only) [detailed version]
H	CITY	City (identifiable cities only)
H	FARM	Farm status
H	NMOTHERS	Number of mothers in household
H	NFATHERS	Number of fathers in household
P	PERNUM	Person number in sample unit
P	PERWT	Person weight
P	VERSIONHIST	Release version for historical full counts
P	HISTID	Consistent historical data person identifier
P	FAMUNIT	Family unit membership
P	FAMSIZE	Number of own family members in household
P	MOMLOC	Mother's location in the household
P	MOMRULE_HIST	Rule for linking mother
P	POPLOC	Father's location in the household
P	POPRULE_HIST	Rule for linking father
P	SPLOC	Spouse's location in household
P	SPRULE_HIST	Rule for linking spouse
P	NCHILD	Number of own children in the household
P	NSIBS	Number of own siblings in household
P	ELDCH	Age of eldest own child in household
P	YNGCH	Age of youngest own child in household
P	RELATE (general)	Relationship to household head [general version]
P	RELATED (detailed)	Relationship to household head [detailed version]
P	SEX	Sex
P	AGE	Age
P	BIRTHYR	Year of birth
P	RACE (general)	Race [general version]
P	RACED (detailed)	Race [detailed version]
P	BPL (general)	Birthplace [general version]
P	BPLD (detailed)	Birthplace [detailed version]
P	NAMELAST	Last name
P	NAMEFIRST	First name

Selected Columns in My DBs

```
TARGET_COLUMNS =[
    YEAR ,
    SAMPLE ,
    SERIAL ,
    PERNUM
    NUMPREC ,
    HHTYPE ,
    TATEICP ,
    OUNTYICP ,
    CITY ,
    FARM ,
    NMOTHERS ,
    NFATHERS ,
    FAMUNIT ,
    FAMSIZE ,
    OMLOC
    OMRULE_HIST ,
    POPLOC ,
    POPRULE_HIST ,
    SPLOC ,
    SPRULE_HIST ,
    NCHILD ,
    NSIBS ,
    ELDCH ,
    YNGCH ,
    RELATED ,
    SEX ,
    AGE ,
    BIRTHYR ,
    RACED ,
    BPLD,
    NAMELAST ,
    NAMEFRST ,
    HISTID
```

```
]
```

I'm Andy Askey : My Goals with this Project

The Erskine/Askey family began around 1775 when Thomas Erskine was awarded many acres of land in central Pennsylvania by the new US government for serving as an officer during the revolutionary war. The name was change to Askey in government records and the family could never reclaim Erskine. (except Uncle William who reclaimed Erskine and moved to California where his descendants still live).

Many of us have been using ancestry in various formats to track family history and talking with some of my California cousins, I think we all kind of got a lot of it right 250 -300 years later.

But a lot of ancestry trees are wrong. I started census only research probably about 20 years ago and did it manually. I found so many errors in other people's trees. My tree has 1500 people and it's only based on the census. That was the manual census. So now I'd like to validate my work with the real census data.

The purpose of this Python CSV project is to validate my family tree. It is in a GitHub project. Anyone else who wants to download the code and do the same thing for themselves, good luck with it. It is free and I ask that you keep it free if you use it for yourself and maybe add to it.

<https://github.com/AJAskey/Genealogy>

Individual Report for Thomas Askey

Individual Summary	Thomas Askey 25 Jul 1727 - 24 Nov 1807
Sex:	Male
Father:	Thomas Aspy Erskine
Mother:	Ellenor Stephenson
Individual Facts	
Birth:	25 Jul 1727 in Antrim, Antrim, Northern Ireland
Immigration:	1742 in Maryland, USA
Residence:	1750 in Maryland, USA
Will:	08 Aug 1755 in Centre, Pennsylvania, USA; Named in will of Thomas Erskine' Aspy' as Thomas ' Aspy' in Chester County, but record is actually from Cumberland County. He is named as executor with Ellenor, and he is oldest son.
Residence:	1760 in Franklin, Fayette, Pennsylvania, USA
Residence:	1770 in Franklin, Fayette, Pennsylvania, USA
Property:	1774 in Howard, Centre, Pennsylvania, USA; Built his homestead on 288 acres on the north bank of Bald Eagle Creek
Military:	Bet. 1775-1783 in Franklin, Fayette, Pennsylvania, USA
Residence:	1780 in Franklin, Fayette, Pennsylvania, USA
Residence:	1790 in Bullskin, Fayette, Pennsylvania, USA
Residence:	1800 in Shippensburg, Cumberland, Pennsylvania, USA
Death:	24 Nov 1807 in Howard, Centre, Pennsylvania, USA
Burial:	28 Nov 1807 in Jacksonville, Centre, Pennsylvania, USA; Lick Run Presbyterian Cemetary
Title:	Captain
Shared Facts	Elizabeth Baker
Marriage:	12 Jun 1764 in Carlisle, Cumberland, Pennsylvania, USA; Rev. John Conrad Bucher
Children:	David Thomas Askey Samuel A Askey Margaret Ellen Askey William Erskine Catherine Askey Robert Alexander Askey Rebecca Askey James Askey John Ellis Askey Anne Askey
Notes	
Person Notes	Birth: 1727 in County Antrim, Northern Ireland Death: 24 NOV 1807 in Howard, Centre County, PA Burial: 1807 Lick Run Presbyterian Church Cemetery, Jacksonville, Centre County, PA Immigration: BET. 1745 - 1750 from County Antrim, Northern Ireland. Military Service: 1760 French and Indian Wars, Lieutenant Military Service: Maryland Revolutionary Records; pension claims/ms, bounty land applications, marriages of soldiers by Harry Wright Newman, Washington, D.C., 1938 Military Service: BET. 1777 - 1782 Revolutionary War, Captain, 1st Class, 1st Co., 1st Btn., Cumberland County Militia. Revolutionary War Soldier "Askey, Capt. Thomas. Died 1807. First marching company, first call July 31, 1777;

DatabaseVault

Function: Builds the foundational SQLite databases.

Details: Loads the CSV census data into the immutable "Raw DB" Vault on my local NVMe drive, keeping the ground truth safe from any matching error.

Origin ID must be at the PERSON level and a composite of (SAMPLE + SERIAL + PERNUM + ID integer). Households split and merge across decades. The incremental ID points to the line in the CSV file if one ever needs to go back and reference where the data came from. SAMPLE identifies the IPUMS sample from which the case is drawn. Each Sample receives a unique 6-digit code. The codes are structured as follows: The first four digits are the year of the census/survey. The next two digits identify the sample within the year.

`composite_id = sample_serial_pernum}_{count}`

This script can run threaded with 1 to n workers. Output can be to one large DB or several smaller DBs.

3 or 4 workers is the best combo on my super-computer.

OS Name	Microsoft Windows 11 Pro
System Name	ELLA5
System Manufacturer	ASUS
Processor Intel Core	Ultra 9 285K, 3700 Mhz, 24 Core(s), 24 Logical Processor(s)
BaseBoard Product	ProArt Z890-CREATOR WIFI
Installed RAM	128 GB
Video	AMD Radeon RX 9070 XT
NVMe disk	Samsung SSD 9100 PRO 4TB

Familiarize.

Function: The core matching engine.

Details: Queries the database using demographic scoring (Age, Household Vectors, and Geography) to link family members across decades without relying on names.

The first pass finds all records associated with the main person. The next pass includes all the family members. A second database was created with lists, which shows the origin ID of all records tied to a family.

BuildGedcom.

Function: Prepares data for Ancestry.com.

Details: Converts the linked, anonymous family records into a standard GEDCOM file format ready for upload.

AuxProcessing.

Function: The bi-directional merge.

Details: Reads the name-enriched data exported back from existing Ancestry.com trees and merges it into your local "Clean DB" to permanently attach names to your St. Joe's IDs.

Family Tree Maker generates various outputs to ingest and merge with census data in the DB.

The merged data is put back into the system via the new GEDCOM created.